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Association of the COVID-19 pandemic on stroke admissions and treatment globally: a systematic review

Rachel A Van Dusen, Kiera Abernethy, Nagendra Chaudhary, Vibhu Paudyal, [Om Kurmi](#)

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Abstract

Objectives: The COVID-19 pandemic has highlighted insufficiencies and gaps within healthcare systems globally. In most countries, including high-

income countries, healthcare facilities were over-run and occupied with too few resources beyond capacity. We carried out a systematic review with a primary aim to identify the influence of the COVID-19 pandemic on the presentation and treatment of stroke globally in populations ≥ 65 years of age. Design: A systematic review was completed. In total, 38 papers were included following full-text screening. Data sources: PubMed, MEDLINE and Embase. Eligibility criteria: Eligible studies included observational and real-world evidence publications with a population who have experienced stroke treatment during the COVID-19 pandemic. Exclusion criteria included studies comparing the effect of the COVID-19 infection on stroke treatment and outcomes. Data extraction and synthesis: Primary outcome measures extracted were the number of admissions, treatment times and patient outcome. Secondary outcomes were severity on admission, population risk factors and destination on discharge. No meta-analysis was performed. Results: This review demonstrated that 84% of studies reported decreased admissions rates during the COVID-19 pandemic. However, among those admitted, on average, had higher severity of stroke. Additionally, in-hospital stroke treatment pathways were affected by the implementation of COVID-19 protocols, which resulted in increased treatment times in 60% of studies and increased in-hospital mortality in 82% of studies by 100% on average. The prevalence of stroke subtype (ischaemic or haemorrhagic) and primary treatment methods (thrombectomy or thrombolysis) did not vary due to the COVID-19 pandemic. Conclusions: During the COVID-19 pandemic, many populations hesitated to seek medical attention, decreasing hospital admissions for less severe strokes and increasing hospitalisation of more severe cases and mortality. The effect of the pandemic on society and healthcare systems needs to be addressed to improve stroke treatment pathways and prepare for potential future epidemics. PROSPERO registration number: CRD42021248564.

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Keywords

- Cardiovascular medicine
- 1506
- 2474
- 1683
- Stroke
- COVID-19
- EPIDEMIOLOGY
- Stroke medicine
- GERIATRIC MEDICINE

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